# Supplementary Material

Systematic benchmarking of functional module partitioning
in GO pathway subnetworks

Supplementary Tables S1--S6

Supplementary Figures S1--S2

Total evaluation records: 3100 (main) + 528 (synthetic) + 317 (downstream)

## Table S1. Cohort characteristics and pathway coverage

| Accession | Disease | Median pathway genes | Samples | Pathways | Algorithm evaluations |
|---|---|---|---|---|---|
| GSE10072 | Lung adenocarcinoma | 91 | 107 | 7 | 7 |
| GSE137342 | Severe sepsis | 124 | 57 | 6 | 6 |
| GSE15653 | Type 2 diabetes / obesity | 117 | 18 | 6 | 6 |
| GSE17536 | Colorectal cancer | 125 | 177 | 7 | 7 |
| GSE19188 | Non-small cell lung cancer | 137 | 156 | 7 | 7 |
| GSE21942 | Multiple sclerosis | 150 | 29 | 6 | 6 |
| GSE25066 | Breast cancer | 115 | 486 | 7 | 7 |
| GSE30153 | Systemic lupus erythematosus | 112 | 26 | 6 | 6 |
| GSE36961 | Hypertrophic cardiomyopathy | 172 | 145 | 7 | 7 |
| GSE38642 | Type 2 diabetes | 121 | 63 | 6 | 6 |
| GSE39088 | Systemic lupus erythematosus | 218 | 142 | 7 | 7 |
| GSE39582 | Colorectal cancer | 102 | 560 | 6 | 6 |
| GSE42568 | Breast cancer | 86 | 121 | 6 | 6 |
| GSE55098 | Type 1 diabetes | 152 | 22 | 6 | 6 |
| GSE61635 | Systemic lupus erythematosus | 31 | 129 | 4 | 4 |
| GSE6764 | Hepatocellular carcinoma | 101 | 72 | 6 | 6 |

## Table S2. Complete algorithm performance metrics

| Algorithm | Modules | Modularity Q | Coherence z | Stability | Robustness | Antagonistic | n |
|---|---|---|---|---|---|---|---|
| CPM γ=0.20 | 140.440 | 0.029 | 3.769 | 0.904 | 0.874 | 0.994 | 100 |
| CPM γ=0.10 | 111.830 | 0.093 | 2.325 | 0.873 | 0.919 | 0.968 | 100 |
| Clique Percolation | 46.500 | 0.046 | 2.225 | 0.915 | 0.934 | 0.295 | 100 |
| CPM γ=0.05 | 85.920 | 0.151 | 1.755 | 0.867 | 0.951 | 0.874 | 100 |
| K-means | 5.620 | 0.140 | 1.397 | 0.739 | 0.874 | 0.613 | 100 |
| Ward | 5.620 | 0.141 | 1.391 | 0.663 | 0.836 | 0.589 | 100 |
| OPTICS | 12.360 | 0.145 | 1.130 | 0.286 | 0.630 | 0.821 | 100 |
| Gaussian Mixture | 5.620 | 0.140 | 1.128 | 0.654 | 0.772 | 0.549 | 100 |
| DBSCAN | 3.930 | 0.087 | 1.046 | 0.527 | 0.858 | 0.254 | 100 |
| HDBSCAN | 4.300 | 0.177 | 0.998 | 0.393 | 0.736 | 0.430 | 100 |
| Walktrap | 9.010 | 0.256 | 0.909 | 0.590 | 0.802 | 0.536 | 100 |
| Leiden γ=1.0 | 4.610 | 0.290 | 0.906 | 0.659 | 0.885 | 0.541 | 100 |
| Louvain γ=0.5 | 16.880 | 0.239 | 0.889 | 0.500 | 0.851 | 0.617 | 100 |
| Louvain | 4.680 | 0.290 | 0.866 | 0.644 | 0.891 | 0.547 | 100 |
| Infomap | 5.660 | 0.237 | 0.863 | 0.743 | 0.942 | 0.394 | 100 |
| Fast Greedy | 4.510 | 0.285 | 0.851 | 0.655 | 0.910 | 0.538 | 100 |
| Label Propagation | 2.680 | 0.221 | 0.831 | 0.677 | 0.911 | 0.098 | 100 |
| Multilevel | 4.630 | 0.289 | 0.820 | 0.646 | 0.841 | 0.590 | 100 |
| Leiden γ=10.0 | 92.300 | 0.040 | 0.802 | 0.663 | 0.891 | 0.981 | 100 |
| Leiden γ=2.0 | 26.910 | 0.200 | 0.796 | 0.646 | 0.875 | 0.793 | 100 |
| Louvain γ=2.0 | 26.210 | 0.201 | 0.789 | 0.628 | 0.855 | 0.821 | 100 |
| Affinity Propagation | 19.430 | 0.221 | 0.727 | 0.683 | 0.820 | 0.847 | 100 |
| Spectral | 5.620 | 0.202 | 0.710 | 0.598 | 0.865 | 0.550 | 100 |
| Leiden γ=0.5 | 2.250 | 0.242 | 0.683 | 0.681 | 0.894 | 0.169 | 100 |
| NMF | 5.590 | 0.183 | 0.667 | 0.501 | 0.790 | 0.681 | 100 |
| Leading Eigenvector | 3.990 | 0.266 | 0.659 | 0.596 | 0.909 | 0.509 | 100 |
| Leiden γ=5.0 | 68.690 | 0.086 | 0.657 | 0.663 | 0.893 | 0.923 | 100 |
| Mean Shift | 6.200 | 0.018 | 0.426 | 0.549 | 0.899 | 0.109 | 100 |
| WGCNA-style Hierarchical | 3.000 | 0.003 | 0.033 | 0.613 | 0.892 | 0.000 | 100 |
| Leiden γ=0.1 | 1.110 | 0.127 | 0.031 | 0.929 | 0.969 | 0.009 | 100 |
| Spin Glass | 1.000 | — | -0.086 | 1.000 | 1.000 | 0.000 | 100 |

## Table S3. Paired statistical tests

| Metric | Pairs | Median (exp) | Median (ref) | p | q (FDR) | Cliff δ | Sig. |
|---|---|---|---|---|---|---|---|
| 模块数 | 100 | 4.500 | 3.000 | 1.95e-16 | 4.87e-16 | +0.870 | yes |
| 模块度Q | 100 | 0.291 | 0.000 | 3.90e-18 | 1.95e-17 | +1.000 | yes |
| 功能内聚z | 100 | 0.689 | 0.045 | 2.74e-07 | 4.57e-07 | +0.418 | yes |
| 稳定性 | 100 | 0.646 | 0.626 | 6.79e-02 | 6.79e-02 | +0.133 | no |
| 拮抗拆分率 | 44 | 0.500 | 0.000 | 5.59e-07 | 6.98e-07 | +0.727 | yes |

Experimental: Leiden γ=1.0. Reference: WGCNA-style hierarchical. Wilcoxon signed-rank, Benjamini-Hochberg correction.

## Table S4. Synthetic benchmark results (ground truth available)

| Algorithm | LFR mean ARI | LFR max ARI | Nested sub-group ARI | Modules |
|---|---|---|---|---|
| CPM γ=0.10 | 0.379 | 1.000 | 1.000 | 12.0 |
| Infomap | 0.000 | 0.000 | 1.000 | 12.0 |
| Label Propagation | 0.237 | 1.000 | 1.000 | 12.0 |
| Leiden γ=2.0 | 0.603 | 0.992 | 1.000 | 12.0 |
| Leiden γ=5.0 | 0.122 | 0.220 | 1.000 | 12.0 |
| Leiden γ=1.0 | 0.927 | 1.000 | 0.943 | 11.3 |
| Louvain | 0.941 | 1.000 | 0.943 | 11.3 |
| Walktrap | 0.774 | 1.000 | 0.943 | 11.3 |
| WGCNA-style Hierarchical | 0.929 | 1.000 | 0.908 | 11.0 |
| CPM γ=0.05 | 0.000 | 0.000 | 0.863 | 10.3 |
| Leiden γ=0.5 | 0.000 | 0.000 | 0.602 | 6.0 |
| Spectral | 0.876 | 1.000 | 0.602 | 6.0 |
| Ward | 0.724 | 1.000 | 0.551 | 6.0 |
| K-means | 0.641 | 1.000 | 0.377 | 6.0 |
| Gaussian Mixture | 0.598 | 1.000 | 0.352 | 6.0 |

LFR: block model across fuzziness $\mu \in \{0.10 – 0.70\}$. Nested: recovery of 12 ground-truth sub-groups (the decisive benchmark).

## Table S5. Null-model comparison

| Cohort | Pathway | Observed Q | Null A (rewired) | Null B (permuted) | z (A) | z (B) |
|---|---|---|---|---|---|---|
| GSE137342 | apoptotic process (GO:0006915) | 0.173 | 0.170 | 0.419 | 0.62 | -12.42 |
| GSE137342 | cellular response to oxidative stress (G | 0.165 | 0.215 | 0.510 | -7.21 | -11.96 |
| GSE137342 | chemokine-mediated signaling pathway (GO | 0.315 | 0.544 | 0.583 | -4.39 | -5.59 |
| GSE137342 | inflammatory response (GO:0006954) | 0.295 | 0.226 | 0.456 | 10.09 | -6.25 |
| GSE137342 | regulation of I-kappaB kinase/NF-kappaB | 0.171 | 0.147 | 0.417 | 6.52 | -10.18 |
| GSE137342 | regulation of immune response (GO:005077 | 0.304 | 0.280 | 0.460 | 1.95 | -6.56 |
| GSE137342 | response to cytokine (GO:0034097) | 0.165 | 0.205 | 0.470 | -6.22 | -13.20 |
| GSE25066 | apoptotic process (GO:0006915) | 0.366 | 0.785 | 0.482 | -12.74 | -3.94 |
| GSE25066 | cellular response to oxidative stress (G | 0.213 | 0.704 | 0.537 | -6.31 | -8.46 |
| GSE25066 | chemokine-mediated signaling pathway (GO | 0.344 | 0.648 | 0.566 | -3.32 | -5.02 |
| GSE25066 | inflammatory response (GO:0006954) | 0.294 | 0.644 | 0.432 | -15.51 | -6.82 |
| GSE25066 | regulation of I-kappaB kinase/NF-kappaB | 0.268 | 0.727 | 0.485 | -14.55 | -8.37 |
| GSE25066 | regulation of immune response (GO:005077 | 0.308 | 0.501 | 0.474 | -7.44 | -6.39 |
| GSE25066 | response to cytokine (GO:0034097) | 0.485 | 0.717 | 0.517 | -7.48 | -0.90 |
| GSE39582 | apoptotic process (GO:0006915) | 0.544 | 0.706 | 0.436 | -6.03 | 4.70 |
| GSE39582 | cellular response to oxidative stress (G | 0.500 | 0.730 | 0.510 | -5.86 | -0.34 |
| GSE39582 | chemokine-mediated signaling pathway (GO | 0.529 | 0.621 | 0.561 | -1.44 | -0.71 |
| GSE39582 | inflammatory response (GO:0006954) | 0.182 | 0.453 | 0.437 | -15.07 | -8.41 |
| GSE39582 | regulation of I-kappaB kinase/NF-kappaB | 0.490 | 0.637 | 0.436 | -7.21 | 2.59 |
| GSE39582 | regulation of immune response (GO:005077 | 0.263 | 0.394 | 0.474 | -6.26 | -7.27 |
| GSE39582 | response to cytokine (GO:0034097) | 0.483 | 0.674 | 0.476 | -6.80 | 0.26 |
| GSE6764 | apoptotic process (GO:0006915) | 0.286 | 0.448 | 0.428 | -8.59 | -7.51 |
| GSE6764 | cellular response to oxidative stress (G | 0.324 | 0.542 | 0.477 | -6.70 | -6.44 |
| GSE6764 | chemokine-mediated signaling pathway (GO | 0.470 | 0.624 | 0.558 | -2.96 | -1.60 |
| GSE6764 | regulation of I-kappaB kinase/NF-kappaB | 0.389 | 0.474 | 0.419 | -5.51 | -1.66 |
| GSE6764 | regulation of immune response (GO:005077 | 0.404 | 0.439 | 0.479 | -1.72 | -3.00 |
| GSE6764 | regulation of inflammatory response (GO: | 0.335 | 0.459 | 0.459 | -11.54 | -4.71 |
| GSE6764 | response to cytokine (GO:0034097) | 0.412 | 0.522 | 0.466 | -3.45 | -2.43 |

These z-scores are computed at the 75th-percentile edge-weight threshold and are highly threshold-dependent (see Table S6): the sign of the comparison reverses across the 50th-95th percentile range. Modularity comparisons against null models should therefore not be reported as a robust property of the data.

## Table S6. Modularity is unstable with respect to graph construction

| Edge-weight threshold | Q (weighted) | Modules | Q (uniform weights) | Modules | Weight contribution |
|---|---|---|---|---|---|
| 50th | 0.269 | 5.5 | 0.097 | 4.5 | 64% |
| 75th | 0.297 | 14.0 | 0.184 | 12.6 | 38% |
| 85th | 0.329 | 23.4 | 0.246 | 22.8 | 25% |
| 90th | 0.362 | 30.1 | 0.292 | 29.5 | 19% |
| 95th | 0.419 | 40.9 | 0.366 | 40.9 | 12% |

Mean Q rises from 0.269 to 0.419 (+56%) across the threshold range, while the comparison against a degree-preserving null model reverses sign. Replacing all edge weights with a constant reduces mean Q from 0.335 to 0.237, i.e. weights contribute 29% of modularity independently of topology.

## Supplementary Figures

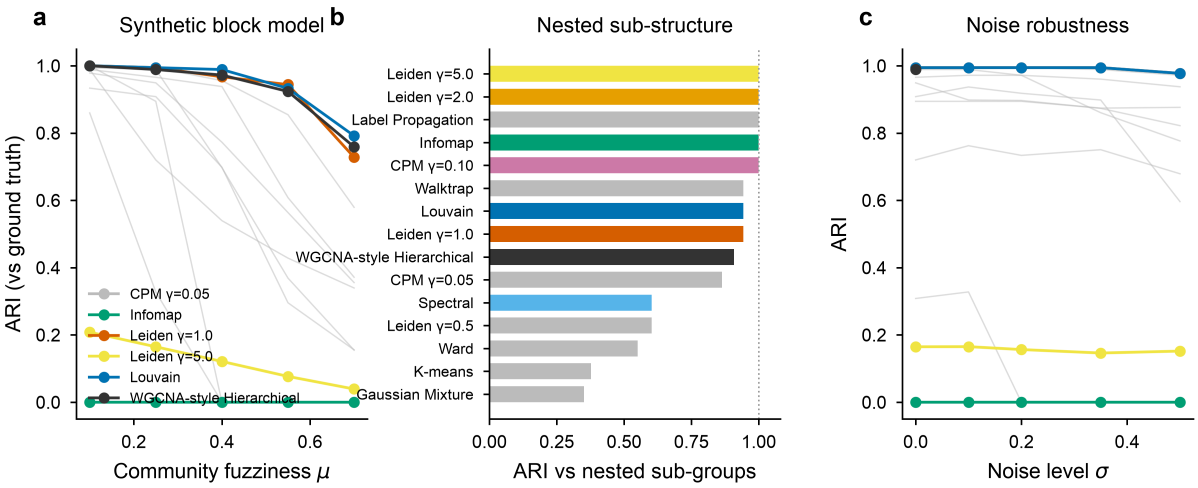



Figure S1. Synthetic benchmark accuracy across fuzziness and noise levels.

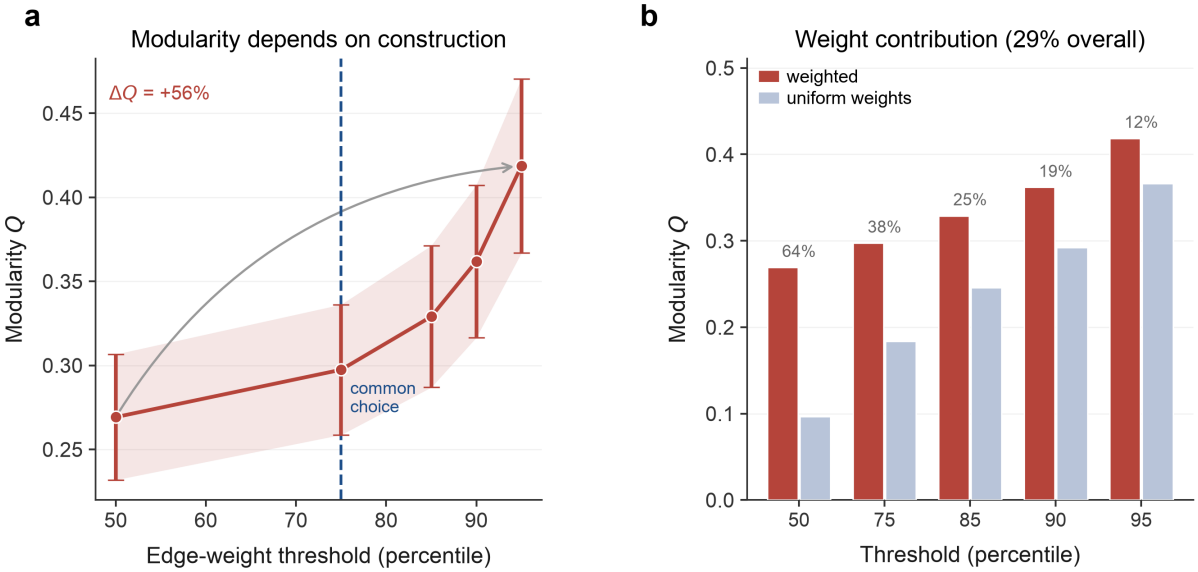



Figure S2. Modularity is unstable with respect to graph construction. Left: mean modularity rises monotonically with the edge-weight threshold. Right: weighted versus uniform-weight modularity, showing the 29% contribution of weights independently of topology.